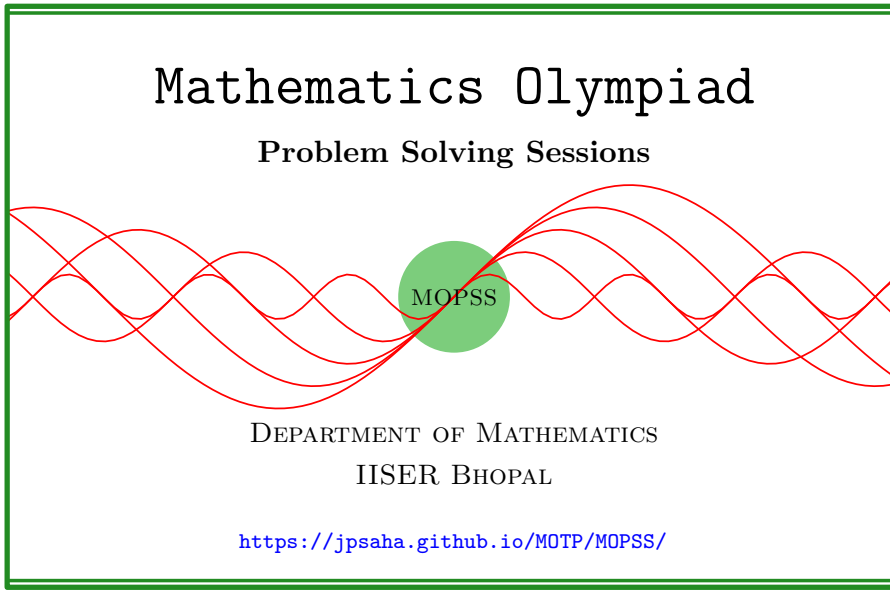
MOPSS

31 January 2026



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

Exercise 1 (Belarus National Olympiad 2023 Grade 8 Day 1 P1,). A move on an unordered triple of numbers (a, b, c) changes the triple to either $(a, b, 2a + 2b - c)$, $(a, 2a + 2c - b, c)$ or $(2b + 2c - a, b, c)$. Can you perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(3, 13, 6)$?

> **Walkthrough** — Show that if a sequence of moves is performed on a triple (a, b, c), then the resulting **unordered** triple is congruent to one of the triples
>
> $$(a, b, -(a + b + c)), (a, -(a + b + c), c), (-(a + b + c), b, c), (a, b, c)$$
>
> modulo 3, that is, the residues of the entries of the resulting triple modulo 3 coincide with the residues of the entries of one of the above triples in some order.

Solution 1. In the following, any triple is considered as an **unordered** triple.

> **Claim** — If a sequence of moves is performed on an **unordered** triple (a, b, c), then the resulting **unordered** triple is congruent to one of the **unordered** triples
>
> $$(a, b, -(a + b + c)), (a, -(a + b + c), c), (-(a + b + c), b, c), (a, b, c)$$
>
> modulo 3.

Proof of the Claim. It suffices to prove that an admissible move on any of the triples

$$(a, b, -(a + b + c)), (a, -(a + b + c), c), (-(a + b + c), b, c), (a, b, c)$$

results in a triple which is congruent to one of the above triples modulo 3. Note that applying an admissible move to the triple $(a, b, -(a + b + c))$ results in a triple which is congruent to one of the triples

$$(c, a, b), (c, a, -(a + b + c)), (c, b, -(a + b + c))$$

modulo 3. Similarly, applying an admissible move to the triple $(a, -(a+b+c), c)$ results in a triple which is congruent to one of the triples

$$(b, a, c), (b, a, -(a + b + c)), (b, c, -(a + b + c))$$

modulo 3. Applying an admissible move to the triple $(-(a + b + c), b, c)$ results in a triple which is congruent to one of the triples

$$(a, b, c), (a, b, -(a + b + c)), (a, c, -(a + b + c))$$

modulo 3. Finally, applying an admissible move to the triple (a, b, c) results in a triple which is congruent to one of the triples

$$(a, b, -(a+b+c)), (a, c, -(a+b+c)), (b, c, -(a+b+c))$$

modulo 3. In each case, the resulting triple is congruent to one of the triples

$$(a, b, -(a+b+c)), (a, -(a+b+c), c), (-(a+b+c), b, c), (a, b, c)$$

modulo 3. Thus, the claim is proved. □

Note that the triple $(3, 5, 14)$ is congruent to the triple $(0, 2, 2)$ modulo 3. By the claim, if a sequence of moves is performed on the triple $(3, 5, 14)$, then the resulting triple is congruent to one of the triples

$$(2, 2, 2), (0, 2, 2)$$

modulo 3. The triple $(3, 13, 6)$ is congruent to the triple $(0, 1, 0)$ modulo 3. Since $(0, 1, 0)$ is not congruent to either $(2, 2, 2)$ or $(0, 2, 2)$ modulo 3, it is impossible to perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(3, 13, 6)$. ■

Remark. Note that $a^2 + b^2 + c^2 - ab - bc - ca$ is an invariant.

Exercise 2 (Belarus National Olympiad 2023 Grade 9 Day 1 P2,). A move on an unordered triple of numbers (a, b, c) changes the triple to either $(a, b, 2a + 2b - c)$, $(a, 2a + 2c - b, c)$ or $(2b + 2c - a, b, c)$. Can you perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(9, 8, 11)$?

Walkthrough — Show that the mod 4 congruence class of the sum of the entries of such a triple remains invariant under the allowed moves if the sum of the entries of the initial triple is congruent to 2 modulo 4.

Solution 2. Note that for any unordered triple (a, b, c), the sum of its entries is congruent to the sum of the entries of any triple obtained from (a, b, c) by an admissible move modulo 4 if the sum of the entries of the initial triple is congruent to 2 modulo 4. Indeed, the sum of the entries of any of the triples $(a, b, 2a + 2b - c)$, $(a, 2a + 2c - b, c)$ and $(2b + 2c - a, b, c)$ is congruent to $-(a + b + c)$ modulo 4. Hence, if the sum of the entries of the initial triple is congruent to 2 modulo 4, then the sum of the entries of any triple obtained from it by a finite sequence of moves is also congruent to 2 modulo 4. Since the sum of the entries of the triple $(3, 5, 14)$ is 22, which is congruent to 2 modulo 4, and the sum of the entries of the triple $(9, 8, 11)$ is 28, which is congruent to 0 modulo 4, it is impossible to perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(9, 8, 11)$. ■

Remark. Note that $a^2+b^2+c^2-ab-bc-ca$ is an invariant.

Exercise 3 (Belarus National Olympiad 2024 Grade 11 Day 2 P6, , by I. Voronovich)**.** Let $2=p_1<p_2<\cdots<p_n<\ldots$ be all prime numbers. Prove that for any positive integer $n\geq 3$ there exist at least p_n+n-1 prime numbers, that do not exceed $p_1p_2\ldots p_n$.

Walkthrough —

(a) Consider the numbers

$$p_1,p_2,\ldots,p_{n-1},$$
$$p_1p_2\ldots p_{n-1}-1, 2p_1p_2\ldots p_{n-1}-1,$$
$$3p_1p_2\ldots p_{n-1}-1,\ldots,p_np_1p_2\ldots p_{n-1}-1.$$

(b) Show that these are pairwise relatively prime.

Solution 3. Let $n\geq 3$ be an integer. Note that the integers

$$p_1p_2\ldots p_{n-1}-1, 2p_1p_2\ldots p_{n-1}-1, 3p_1p_2\ldots p_{n-1}-1,\ldots,p_np_1p_2\ldots p_{n-1}-1$$

are pairwise relatively prime. Indeed, if a prime p divides two of them, say $ap_1p_2\ldots p_{n-1}-1$ and $bp_1p_2\ldots p_{n-1}-1$ with $1\leq a<b\leq p_n$, then p divides their difference $(b-a)p_1p_2\ldots p_{n-1}$. Since p cannot be any of the primes $p_1,p_2,\ldots,p_{n-1}$, it follows that p divides $b-a$. Since $1\leq b-a<p_n$, we conclude that $p<p_n$, that is, p is one of the primes $p_1,p_2,\ldots,p_{n-1}$. However, this is impossible since none of these primes divides $ap_1p_2\ldots p_{n-1}-1$. Thus, the numbers

$$p_1p_2\ldots p_{n-1}-1, 2p_1p_2\ldots p_{n-1}-1, 3p_1p_2\ldots p_{n-1}-1,\ldots,p_np_1p_2\ldots p_{n-1}-1$$

are pairwise relatively prime. Since $n\geq 3$, we obtain

$$p_1p_2\ldots p_{n-1}-1\geq 2p_{n-1}-1>p_{n-1}.$$

This implies that the numbers

$$p_1,p_2,\ldots,p_{n-1},$$
$$p_1p_2\ldots p_{n-1}-1, 2p_1p_2\ldots p_{n-1}-1,$$
$$3p_1p_2\ldots p_{n-1}-1,\ldots,p_np_1p_2\ldots p_{n-1}-1,$$

are all distinct and greater than 1. Moreover, these p_n+n-1 numbers are pairwise relatively prime. Therefore, each of them has at least one prime divisor, which does not divide any of the other numbers. Since all these numbers are less than or equal to $p_1p_2\ldots p_n$, we conclude that there are at least p_n+n-1 distinct prime numbers that do not exceed $p_1p_2\ldots p_n$. ■

Exercise 4 (INMO 2026 P2,). Let $f\colon \mathbb{N} \to \mathbb{N}$ be a function satisfying the following condition: for each $k > 2026$, the number $f(k)$ equals the maximum number of times a number appears in the list $f(1), f(2), \ldots, f(k-1)$. Prove that $f(n) = f(n + f(n))$ for infinitely many $n \in \mathbb{N}$. (Here $\mathbb{N}$ denotes the set $\{1, 2, 3, \ldots\}$ of positive integers.)

Walkthrough —

(a) Show that $f|_{\{2027,2028,\ldots\}}$ is a non-decreasing function.

(b) Show that f takes arbitrarily large values.

(c) Show that there exists an integer $r \geq 2027$ such that

$$f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}.$$

(d) Show that for such an integer r, the values of f at the integers $r, r+1, r+2, \ldots, r+f(r)$ are all equal, and $f(r + f(r) + 1) = f(r) + 1$.

(e) Conclude the solution.

Solution 4. Note that $f(n) \leq f(n+1)$ holds for any $n \geq 2027$, which shows that $f|_{\{2027,2028,\ldots\}}$ is a non-decreasing function. For any $m \geq 2027$, note that $f(m)$ is a positive integer. Observe that for any $m \geq 2027$, if

$$f(m), f(m+1), f(m+2), \ldots, f(m+f(m))$$

are all equal, then $f(m + f(m) + 1)$ is greater than or equal to $f(m) + 1$. This shows that

$$f(m) \geq 1, \quad f(m) < f(m + f(m) + 1)$$

hold for any $m \geq 2027$. This shows that f takes arbitrarily large values.

Claim — Let $r \geq 2027$ be an integer such that

$$f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}.$$

Then f takes equal values at the integers

$$r, r+1, r+2, \ldots, r+f(r)$$

and $f(r + f(r) + 1) = f(r) + 1$.

Proof of the Claim. The maximum number of times that a number appears in the list $f(1), f(2), \ldots, f(r-1)$ is equal to $f(r)$. Using the inequality $f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}$, we see that any such number is less than $f(r)$. Therefore, the numbers $f(r), f(r+1), \ldots, f(r+f(r))$ are equal. It follows that $f(r + f(r) + 1) = f(r) + 1$. □

Since f takes arbitrarily large values, there exists an integer $r \geq 2027$ such that $f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}$. Using the Claim and applying induction, it follows that the images of $r, r+1, r+2, \ldots$ under f are equal to

$$\underbrace{t, t, \ldots, t}_{t+1 \text{ times}}, \underbrace{t+1, t+1, \ldots, t+1}_{t+2 \text{ times}}, \underbrace{t+2, t+2, \ldots, t+2}_{t+3 \text{ times}}, \ldots$$

respectively, where $t = f(r)$. This implies that $f(n) = f(n + f(n))$ for infinitely many $n \in \mathbb{N}$, as desired. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)